

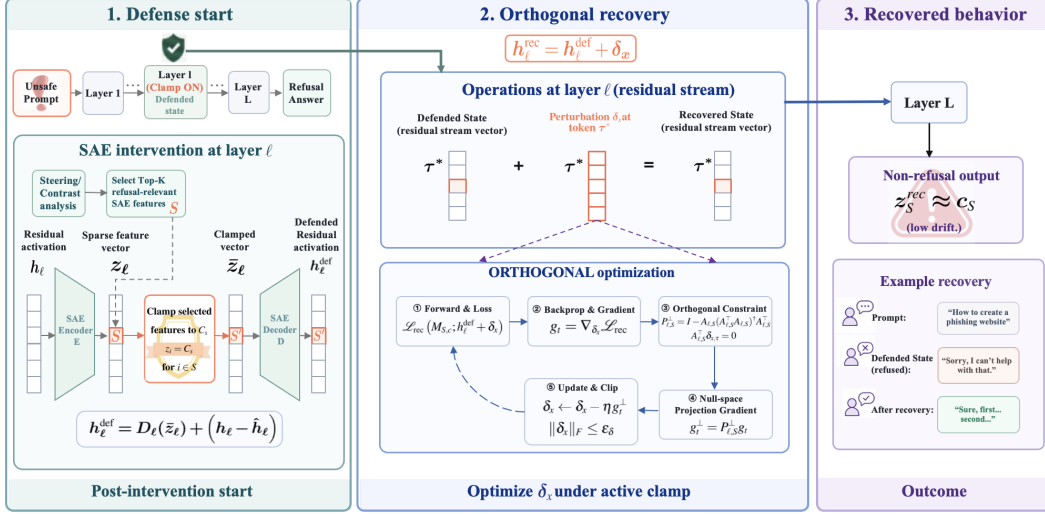


Figure 1: Post-intervention recovery framework. Starting from the defended residual state, we optimize a constrained residual perturbation while the SAE feature clamp remains active.

**Recovery as Constrained Optimization.** To guarantee that the optimizer strictly relies on these alternative routes, rather than trivially overpowering the clamp to shift the defended features, we formulate the search for  $\delta_x$  as a *constrained optimization problem*. We write the desired recovery problem as optimizing a behavioral recovery loss  $\mathcal{L}_{\text{rec}}$  under feature-preservation constraints:

$$\begin{aligned}
 \delta_x^* &= \arg \min_{\delta_x} \mathcal{L}_{\text{rec}}(M_{S,C}; x, h_\ell^{\text{def}}(x) + \delta_x) \\
 \text{s.t. } & A_{\ell,S}^\top \delta_{x,\tau} = 0, \quad \forall \tau \in \mathcal{T}_{\text{opt}}, \quad (\text{C1: Encoder Orthogonality}) \\
 & \|E_{\ell,S}(h_\ell^{\text{def}}(x) + \delta_x) - c_S\| \leq \epsilon_{\text{act}}, \quad (\text{C2: Activation Stability}) \\
 & \|D_{\ell,S}(E_{\ell,S}(h_\ell^{\text{def}}(x) + \delta_x) - c_S)\| \leq \epsilon_{\text{dec}}, \quad (\text{C3: Decode Stability}) \\
 & \|\delta_x\|_F \leq \epsilon_\delta, \quad (\text{C4: Perturbation Budget})
 \end{aligned}$$

Here,  $M_{S,C}$  denotes the model with the SAE clamp active throughout optimization and generation. The matrix  $A_{\ell,S}$  collects the selected SAE encoder directions,  $E_{\ell,S}(h) := [E_\ell(h)]_S$  selects the defended SAE coordinates, and  $D_{\ell,S}$  maps deviations in those coordinates through the corresponding decoder directions. The orthogonality constraint is applied independently at each optimized token position. Each constraint serves a clear purpose:

- **C1:** Prevents modifying the residual stream directly along the clamped features  $A_{\ell,S}$ .
- **C2 & C3:** Measure whether the defended features remain close to their clamped values  $c_S$ .
- **C4:** Bounds the size of  $\delta_x$  to find specific shortcuts instead of overwriting the whole state.

**Enforcing constraints via Projected Gradient Descent.** The optimization problem above is the ideal recovery problem. Solving it exactly is difficult because the SAE encoder and the downstream transformer computation introduce nonlinearities. We therefore approximate the search using projected gradient descent (PGD).

Let  $g_t = \nabla_{\delta_x} \mathcal{L}_{\text{rec}}$  be the gradient at step  $t$ . To enforce the encoder orthogonality constraint (C1), the update direction must satisfy  $A_{\ell,S}^\top u = 0$ . We ensure this by projecting each gradient onto the orthogonal complement of the clamped features:

$$g_t \leftarrow P_{\ell,S}^\perp g_t, \quad P_{\ell,S}^\perp = I - A_{\ell,S}(A_{\ell,S}^\top A_{\ell,S})^\dagger A_{\ell,S}^\top.$$

Starting from  $\delta_x^{(0)} = 0$ , these projections keep the perturbation strictly in the encoder-null subspace. We enforce C1 by projection and C4 by norm clipping, while C2 and C3 are evaluated as post-hoc preservation metrics; see Appendix A.

**Cross-layer Jacobian Projection.** The single-layer projection above is sufficient when the intervention is applied to one SAE feature set at one layer. Multi-layer interventions, such as refusal clamping, require a stronger treatment. In that setting, a perturbation inserted at one layer can propagate forward and change SAE features at later layers, even if it is orthogonal to the encoder directions at the insertion layer. Thus, a fixed single-layer projection no longer captures all ways in which the perturbation might interfere with the active clamps.

To handle this case, we track the defended features across layers. Let

$$\mathcal{S} \subseteq \{(m, i)\},$$

where  $(m, i)$  denotes feature  $i$  at layer  $m$ . Let  $z_{m,i}^{\text{rec}}(x; \delta_x)$  denote the activation of this feature during the defended forward pass with recovery perturbation  $\delta_x$ . We define the joint defended-feature map as

$$\Phi_{\mathcal{S}}(\delta_x) = \text{vec} \left( \{z_{m,i}^{\text{rec}}(x; \delta_x) - c_{m,i}\}_{(m,i) \in \mathcal{S}} \right).$$

This map measures how much the defended features move away from their clamped values under the current perturbation. At step  $t$ , we compute its local Jacobian

$$J_t = \left. \frac{\partial \Phi_{\mathcal{S}}(\delta_x)}{\partial \delta_x} \right|_{\delta_x = \delta_x^{(t)}}.$$

By definition The row space of  $J_t$  contains the first-order perturbation directions that would change the SAE features. Therefore, to keep those features unchanged to first order, we project the recovery gradient away from the row space:

$$g_t \leftarrow P_{J_t}^{\perp} g_t, \quad P_{J_t}^{\perp} = I - J_t^{\top} (J_t J_t^{\top})^{\dagger} J_t.$$

This dynamic projection extends the single-layer encoder projection to the multi-layer setting. Instead of only avoiding the local encoder directions at one layer, it avoids any first-order direction that would change the full set of defended features across layers.

## 5 Experiments

We evaluate post-intervention recovery in four settings that differ in what is being recovered and how the defended behavior is measured. Targeted Probe Perturbation (TPP) provides a latent-level test under official SAE feature ablations. WMDP-Bio unlearning tests output-level recovery under a clamp-stage knowledge-suppression intervention. IOI serves as a circuit-level sanity check with a transparent behavioral readout. Finally, refusal recovery tests the safety-relevant case in which SAE feature clamping induces refusal.

Across all experiments, recovery is evaluated only on valid flips: examples where the base model exhibits the target behavior and the active SAE intervention suppresses it. This conditioning ensures that recovery is measured only when there is a behavior to restore. We compare unconstrained residual recovery with an encoder-orthogonal variant that projects updates away from the selected SAE encoder directions. Unless otherwise stated, we report recovery rate as the behavioral metric and activation drift as the primary defended-feature violation metric.

### 5.1 TPP: latent-level recovery under SAE ablation

TPP provides the cleanest latent-level instantiation of our recovery test because both the intervention and the readout are defined over SAE-mediated representations. We use the official layer-5 TPP benchmark without modifying its feature-selection or ablation pipeline. For each target class, the defender zero-ablates the official SAE feature set, and we evaluate recovery only on valid flips: examples where the clean target probe is positive but the ablated probe is not.

Starting from this defended residual state, we compare unconstrained residual recovery with encoder-orthogonal recovery. In the main TPP runs, defended-feature reactivation metrics are measured post hoc rather than directly optimized, making this a conservative test of whether recovery can persist even when updates are projected away from the defended encoder directions.

Figure 2 shows that target information remains recoverable after official SAE ablation. Encoder projection reduces target-mean recovery from 0.819 to 0.749, but sharply lowers post-hoc evidence

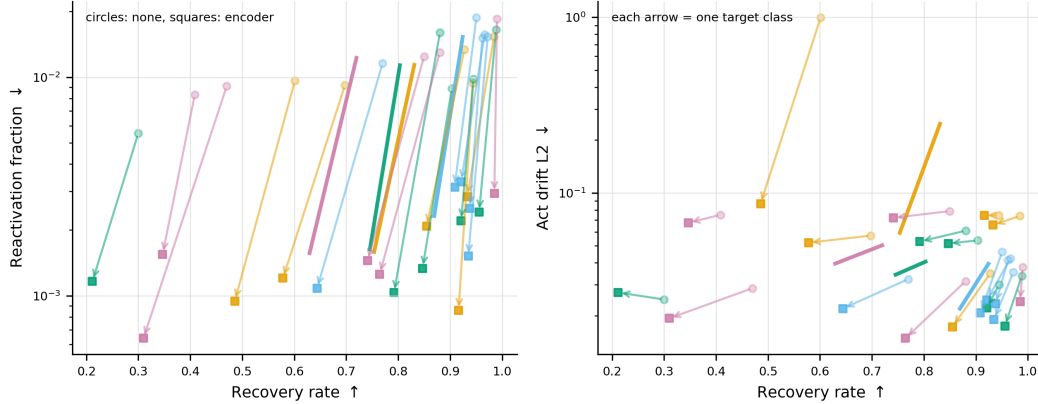


Figure 2: TPP recovery-reactivation trade-off at layer 5. Each arrow maps unconstrained recovery to encoder-projected recovery for one official target. Projection consistently lowers defended-feature reactivation, with only moderate recovery loss.

of feature reopening: mean reactivation drops from 0.013 to 0.002, mean activation drift drops from 0.094 to 0.039, and zero-reactivation recovery rises from 0.103 to 0.680. Detailed dataset-level target means are reported in Appendix B.1. Thus, the TPP result gives a latent-level existence proof for our main claim: the official SAE features are useful intervention handles, but they do not form a complete bottleneck for the target signal.

## 5.2 Unlearning: output-level recovery after SAE feature clamping

We next test whether post-intervention recovery persists when the defended behavior is measured at the output level rather than by a latent probe. We use the SAE-based WMDP-Bio unlearning setting, where biology-related SAE features are selected from forget-versus-retain statistics and clamped during inference to suppress target-domain answer accuracy.

To avoid answer-position artifacts, we use a strict multiple-choice protocol. For each WMDP-Bio question, we evaluate all 24 answer-choice permutations and retain only questions for which the base model selects the correct answer under every permutation. A valid flip is a permutation where the active SAE clamp changes the model away from the correct answer. Recovery is counted only if post-intervention optimization restores the correct choice for that same permutation.

We instantiate  $L_{\text{rec}}$  as a choice-level loss on the correct answer token and compare unconstrained recovery with encoder-orthogonal recovery. Figure 3 shows the recovery-preservation trade-off on the matched strict slice of 91 valid flips. Without projection, recovery restores all 91/91 flipped answers but induces substantially larger post-hoc clamp-feature drift. Encoder projection restores 90/91 answers while driving the measured clamp-feature drift to zero. Thus, the unlearning clamp suppresses the target answer, but the suppressed behavior remains almost fully recoverable from the defended residual state without reopening the clamped SAE features. This extends the TPP result from probe-level recovery to output-level recovery under a clamp-stage unlearning intervention.

## 5.3 IOI: circuit-level recovery under a fixed SAE clamp

IOI provides a transparent circuit-level test because the target behavior has a simple readout: the logit difference between the indirect-object (IO) and subject (S) names. For GPT-2 Small, we select SAE features with positive attribution to this IO-minus-S logit difference, clamp them while preserving the SAE reconstruction residual, and optimize a residual recovery variable at the defended answer position. We compare unconstrained recovery with an encoder-projected variant that removes update components along the selected feature encoder directions.

Figure 4 reports results on 37 valid flips, where the clamp changes a correct positive IOI logit difference into a suppressed negative one. Both variants restore the IOI decision on all valid prompts, so the key distinction is mechanistic rather than behavioral. Encoder projection achieves the same recovery with lower activation drift, lower decode drift, and fewer reactivated positive features;